

Mechanics of Material II

PROJECT

REPORT

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ABSTRACT:

The purpose of this project is to design and analyse a rear dead axle for a Hyundai H-100 pickup truck capable of carrying a maximum load of 1500 kg. The rear dead axle is an important structural component responsible for supporting the vehicle weight, transmitting loads from the wheels to the suspension system, and maintaining wheel alignment. In this report, the axle is designed based on static loading conditions and material strength considerations using the principles of mechanics of materials.

The project includes load estimation, bending moment calculations, shear force analysis, principal stress determination, and optimization of the axle geometry and material selection. A hollow circular axle cross-section is selected to reduce mass while maintaining structural integrity. Mild steel and alloy steel materials are compared based on strength-to-weight ratio, machinability, and cost.

The designed axle is analysed under maximum loading conditions considering both vertical loads and shock loading factors. Principal stresses are calculated using combined stress equations and Mohr's circle concepts. Factor of safety is incorporated to ensure safe operation under practical conditions.

The final optimized design uses alloy steel with a hollow circular cross-section, resulting in reduced weight and improved stress distribution compared to a solid axle. The report concludes with limitations, possible sources of error, and recommendations for future improvements.



Figure 1

INTRODUCTION

A rear axle is one of the most important structural components of a vehicle. It supports the load of the vehicle body and passengers while maintaining wheel alignment and transmitting forces generated during operation. In pickup trucks, the rear axle experiences large bending and shear stresses because of heavy cargo loading.

The Hyundai H-100 pickup is widely used for commercial transportation due to its durability and moderate payload capacity. The rear dead axle used in this vehicle does not transmit torque from the engine but mainly supports vertical loads. Therefore, its design must ensure adequate strength, stiffness, and fatigue resistance.



Figure 2

The design of a rear dead axle involves determining suitable dimensions and materials such that the stresses developed remain below the allowable stress limits. Mechanics of materials principles such as bending stress, shear stress, and principal stress analysis are essential for safe and efficient design.

This report presents the complete design procedure for a rear dead axle capable of carrying a maximum load of 1500 kg. Analytical calculations are performed to evaluate the stresses and

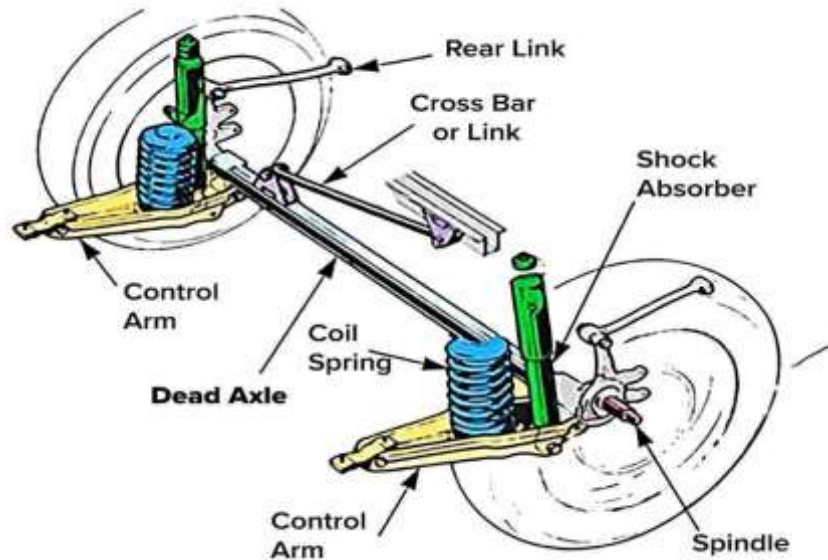


Figure 3

OBJECTIVES:

The major objectives of this project are:

- To design a rear dead axle for a Hyundai H-100 pickup truck.
- To ensure the axle can safely support a maximum load of 1500 kg.
- To calculate bending stress, shear stress, and principal stresses.
- To optimize the axle geometry for minimum weight.
- To select a suitable material based on mechanical properties.
- To apply mechanics of materials concepts in practical engineering design.
- To evaluate the factor of safety for the designed axle.

LITERATURE REVIEW:

Automobile axles are designed based on static and dynamic loading conditions. Researchers have studied different axle configurations including solid and hollow axles to reduce weight while maintaining strength.

Studies show that hollow axles provide better strength-to-weight ratio than solid axles because material near the centre contributes less to bending resistance. Alloy steels such as AISI 4140 are commonly used because of their high yield strength and good fatigue resistance.

Modern design techniques include finite element analysis for stress evaluation, but analytical methods remain important for preliminary design and validation. Mechanics of materials equations are widely used to determine bending stress and principal stresses under combined loading.

REAR DEAD AXLE OVERVIEW:

A **dead axle**, also known as a **non-driven axle**, is a type of axle that does not receive power from the engine. Unlike **live axles**, which actively drive the wheels, a dead axle primarily supports the vehicle's

weight and improves stability. Dead axles are commonly found in trucks, trailers, and some SUVs, playing a crucial role in enhancing load-bearing capacity and ride comfort.

Dead Axle Vs Live Axle:

Understanding the distinction between a dead axle and a live axle is essential in vehicle mechanics.

- **Live Axle:** Transfers power from the engine to the wheels, enabling motion.
- **Dead Axle:** Supports the vehicle's structure but does not contribute to propulsion.

While live axles are used in powered wheels, dead axles are strategically placed in various vehicle types for improved weight distribution and load management

Functions of Rear Dead Axle

- Supports vehicle weight.
- Maintains wheel alignment.
- Resists bending and shear loads.
- Transfers load to suspension system.
- Provides mounting for wheels and brakes.

The axle considered in this project is a rear dead axle subjected mainly to vertical loading.

HYUNDAI H-100 SPECIFICATIONS

The Hyundai H-100 is a light commercial pickup vehicle used for cargo transportation.

Typical Specifications

Parameter	Value
Vehicle Type	Pickup Truck
Maximum Payload	1500 kg
Rear Axle Type	Dead Axle
Number of Rear Wheels	2

For design calculations, the maximum payload is considered as 1500 kg.

DESIGN METHODOLOGY:

The design methodology includes the following steps:

- Determine loading conditions.
- Calculate reactions at wheel supports.
- Determine shear force and bending moment.
- Select suitable material.
- Design axle geometry.
- Calculate stresses.

- Determine principal stresses.
- Optimize geometry and material.
- Verify factor of safety.

ASSUMPTIONS

The following assumptions are made for the design:

- The axle is simply supported at wheel mounting locations.
- The load is pointing load on 0.3m away from the wheels
- Static loading conditions are considered.
- Shock and impact factor of 1.5 is included.
- The axle behaves as a linear elastic material.
- Stress concentration effects are neglected.
- Symmetrical loading is assumed.

LOAD CALCULATIONS:

The load calculations for the rear dead axle are based on the maximum payload capacity of the pickup truck and the distribution of load on the rear axle.

Total Vehicle Load

The total load acting on the vehicle can be calculated using:

The formula is $W = mg$

- Where w is weight
- M is mass
- G is gravitational acceleration.

Main Loads Acting on Axle (65% on dead axle)

$$W_d = 0.65 \times 1500 \times 9.81 = 9564.75 \text{ N}$$

Load on Point C and Point D are equally distributed that is $W_d / 2 = 4782.375 \text{ N}$ each

SHEAR FORCE AND BENDING MOMENT ANALYSIS:

The rear dead axle is modelled as a simply supported beam subjected to vertical loading.

Support Reactions

For symmetrical loading conditions:

$$R_A = R_B = W_d / 2$$

Where:

- R_A and R_B = Reactions at supports

- W_d = Dynamic load acting on axle

Shear Force

The maximum shear force occurs near the supports and is equal to the reaction force.

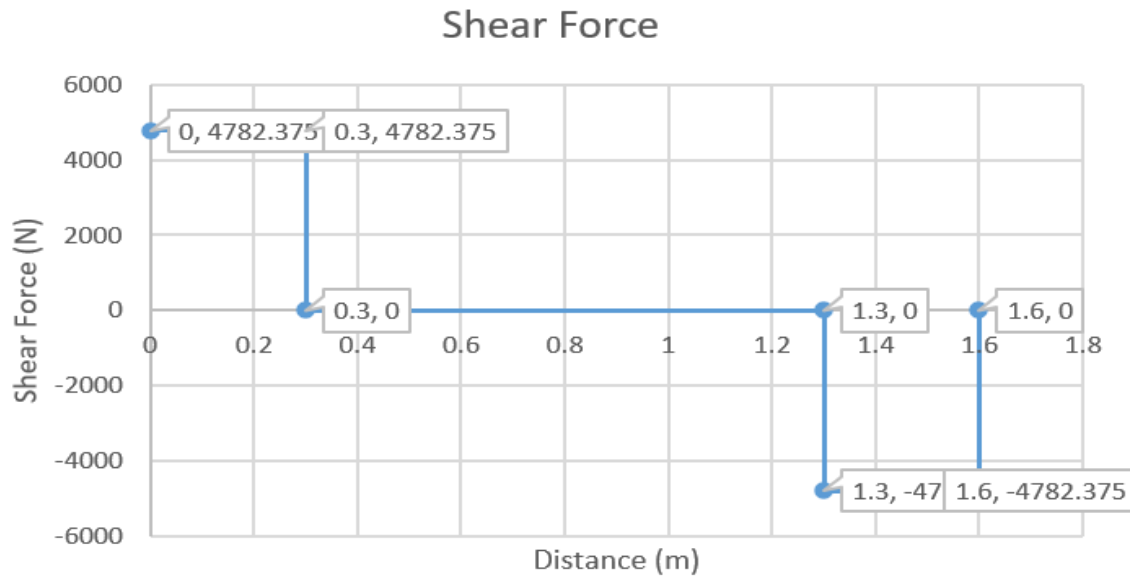


Figure 4

Maximum Bending Moment

Or it can be calculated using bending moment diagram.

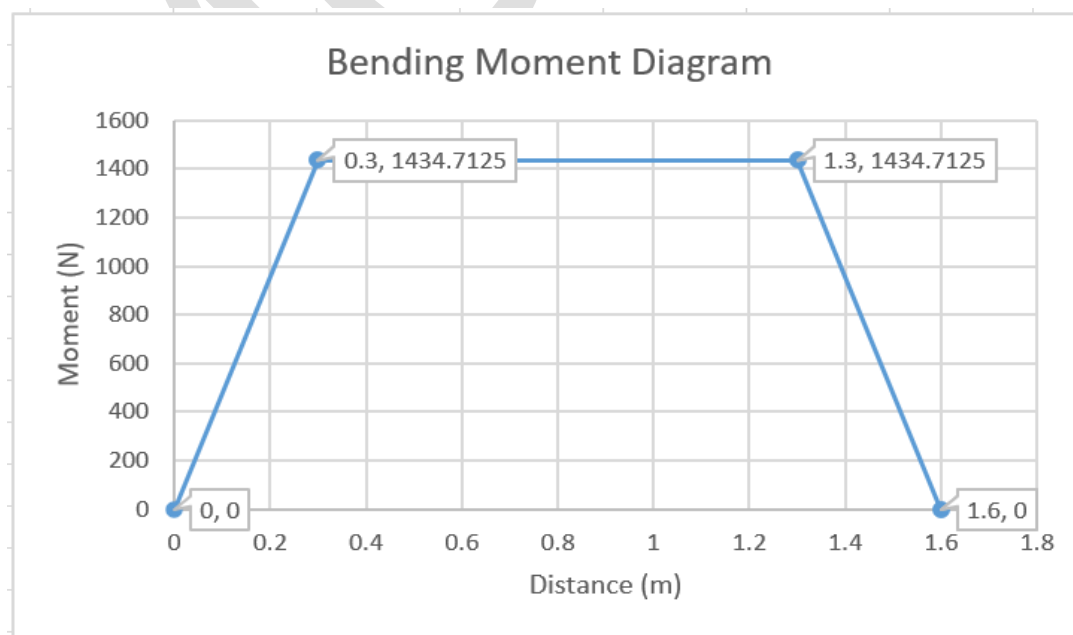


Figure 5

MATERIAL SELECTION

The material must possess:

- High yield strength
- Good fatigue resistance
- Toughness
- Machinability
- Weld ability
- Corrosion resistance

Candidate Materials

Material	Yield Strength (MPa)	Density (kg/m ³)
Mild Steel	250	7850
AISI 1045 Steel	530	7850
AISI 4140 Alloy Steel	415	7850

Selected Material

AISI 4140 alloy steel is selected because:

- High strength-to-weight ratio
- Better fatigue performance
- Suitable for automotive applications
- Allows smaller cross-section size

AXLE GEOMETRY DESIGN

A hollow circular cross-section is selected for the rear axle because it provides high bending resistance with reduced weight.

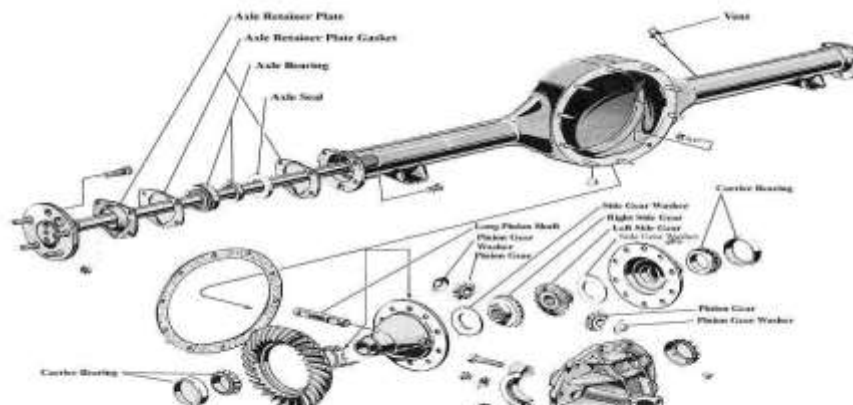


Figure 6

Bending Stress Equation

The bending stress in the axle is determined using:

$$\sigma = Mc / I$$

Where:

- σ = Bending stress
- M = Bending moment
- c = Distance from neutral axis to outer surface
- I = Moment of inertia

Moment of Inertia for Solid Circular Section

$$I = \pi (D^4) / 64$$

Where:

- D = diameter

Allowable Stress

The formula for allowable stress is:

$$\sigma_{allow} = \frac{\sigma_y}{FOS}$$

Where:

- σ_{allow} = Allowable stress
- σ_y = Yield strength of the material
- FOS = Factor of Safety

STRESS ANALYSIS:

The rear dead axle is subjected primarily to:

1. Bending stress
2. Shear stress

Bending Stress

Bending Stress Formula

$$\sigma = \frac{Mc}{I}$$

Where:

- σ = Bending stress
- M = Bending moment
- c = Distance from neutral axis
- I = Moment of inertia of cross-section

Shear Stress

The axle also experiences torsional shear stress due to breaking force

The approximate shear stress equation is:

$$\tau = Tc / J$$

Where:

τ = Shear stress

T = Torque

c = Distance from neutral axis to outer surface

J = Moment of inertia

Cross-sectional Area

For a Solid circular section:

$$A = \pi (D^2) / 4$$

Where:

- D = diameter

The bending stress is generally much higher than the shear stress and therefore dominates the axle design.

Principle stresses:

The rear axle experiences combined bending stress and shear stress. Therefore, principal stresses are determined using stress transformation equations.

The principal stress equation helps determine the maximum stress acting at a point in the axle.

The formula for principal stresses acting on the axle is:

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

where:

- σ_1 = Maximum principal stress
- σ_2 = Minimum principal stress
- σ = Normal (bending) stress
- τ = Shear stress

This equation is used when the axle experiences combined bending and shear loading. It helps determine the maximum stress acting on the axle material and ensures that the design remains within safe stress limits.

Importance of Principal Stress Analysis

Principal stress analysis is important because:

- It predicts the maximum stress condition.
- It helps prevent material failure.
- It ensures safe axle operation.
- It assists in selecting suitable material and geometry.

The calculated principal stresses must remain below the allowable stress of the selected material.

OPTIMIZATION OF DESIGN

Optimization aims to reduce:

- Mass
- Material usage
- Cross-sectional size

while maintaining safe stress limits.

Solid Axle (Design Selected)

Solid Axle

Advantages:

- Simple manufacturing
- Higher stiffness

Disadvantages:

- Higher weight
- More material usage

Optimized Design Parameters:

Parameter	Value
Diameter	45mm
Material	AISI 4140 Steel
Maximum Stress	166.4 MPa
Factor of Safety	2.5

FACTOR OF SAFETY:

The factor of safety ensures that the axle can safely withstand unexpected loads and operating conditions.

The factor of safety is calculated using:

Based on Load (Force)

$$FoS = \frac{\text{Ultimate Load}}{\text{Allowable Load}}$$

A higher factor of safety increase's reliability but may also increase axle size and weight.

The selected factor of safety depends on:

- Vehicle operating conditions
- Type of loading

The formula for principal stresses acting on the axle is:

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

Where:

- σ_1 = Maximum principal stress
- σ_2 = Minimum principal stress
- σ = Normal (bending) stress
- τ = Shear stress

- Material properties
- Manufacturing quality
- Safety requirements

A factor of safety (FOS) of 2.5 is selected for the rear dead axle to ensure safe operation under practical working conditions. The axle is subjected to dynamic and shock loads caused by road bumps, braking, and uneven surfaces, which can produce stresses higher than theoretical values. The selected FOS also accounts for uncertainties in material properties, manufacturing defects, and dimensional inaccuracies. Since the axle experiences repeated loading during vehicle operation, fatigue failure is also an important consideration. Automotive structural components are generally designed with a factor of safety between 2 and 3 for reliable performance. A value of 2.5 provides a good balance between safety, strength, weight reduction, and cost. Therefore, the chosen factor of safety ensures that the axle can withstand unexpected loads without failure while maintaining durability and reliability.

RESULTS AND DISCUSSION:

$V_{\max}$	4782.375 N
$M_{\max}$	1434.7125 Nm
$\sigma_{\text{allowable}}$	166 MPa
$\tau_{\text{allowable}}$	96 MPa
d	45 mm
σ_m	160.4 MPa
F_B	3347.6625 N
T	1004.29875 Nm
τ_m	56.1 MPa
$\sigma_{\max}$	-178.07 MPa
$\sigma_{\min}$	17.67 MPa
$\tau_{\max}$	97.87 MPa
θ_P	35.36°

The designed rear dead axle successfully supports the required load capacity of 1500 kg. Analytical calculations show that the maximum bending stress is within allowable limits.

The use of AISI 4140 alloy steel improves structural performance and allows reduction in axle size compared to mild steel.

The solid axle design provides sufficient strength and stiffness to support the maximum payload of 1500 kg. The design remains within allowable stress limits and ensures reliable operation under service conditions.

The calculated factor of safety indicates reliable operation under practical conditions.

The beam fails at FOS=2.5 first, then we try it at FOS = 2.3 it satisfies all the stresses and now beam is safe at FOS = 2.3, all the stresses both shear and normal stress and at principal stresses.

FOS = 2.3

New Allowable Stresses	
$\sigma_{\text{allowable}}$	180.4 MPa

$\tau_{allowable}$	104.3 MPa
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MANUFACTURING CONSIDERATIONS

The following manufacturing methods can be used:

- Forging
- Machining
- Heat treatment
- Grinding

Heat Treatment

Heat treatment improves:

- Hardness
- Fatigue resistance
- Strength

Processes include:

- Quenching
- Tempering

Surface Finishing

Surface finishing reduces stress concentrations and improves fatigue life.

LIMITATIONS:

The following limitations exist in this project:

- Dynamic loading effects are simplified.
- Fatigue analysis is not performed in detail.
- Stress concentration effects are neglected.
- Real road conditions may produce higher impact loads.
- Temperature effects are not considered.
- Finite element analysis is not included.

SOURCES OF ERROR:

Possible sources of error include:

- Approximation in shock loading factor
- Material property variations
- Dimensional inaccuracies
- Neglecting residual stresses
- Simplified support conditions

- Human calculation errors

CONCLUSION:

In this project, a rear dead axle for a Hyundai H-100 pickup truck was successfully designed using the principles of Mechanics of Materials. The axle was designed to safely support a maximum payload of 1500 kg while maintaining structural integrity under the applied loading conditions. The loading analysis was performed and the maximum bending moment acting on the axle was determined.

AISI 4140 alloy steel was selected as the axle material because of its high strength, toughness, and reliability in automotive applications. A solid circular axle geometry was adopted to provide sufficient strength and stiffness for the expected service conditions. Stress analysis indicated that the maximum bending stress, shear stress, and principal stresses remained within the allowable limits of the selected material.

The final design achieved:

- Safe operation under the specified loading conditions
- High structural strength and rigidity
- Reliable performance with reduced risk of failure
- Acceptable factor of safety for automotive applications

Overall, the project demonstrates the successful application of mechanics of materials concepts in the design of an automotive rear dead axle. The results confirm that the designed solid axle can safely withstand the imposed loads and meet the functional requirements of the Hyundai H-100 pickup truck.

REFERENCES:

1. Gere, J. M., Mechanics of Materials, Cengage Learning.
2. Beer, F. P., Johnston, E. R., Mechanics of Materials, McGraw Hill.
3. Khurmi, R. S., Machine Design, S. Chand Publications.
4. Bhandari, V. B., Design of Machine Elements, McGraw Hill.
5. Hyundai H-100 Technical Specifications Manual.
6. Shigley, J. E., Mechanical Engineering Design, McGraw Hill.
7. Norton, R. L., Machine Design: An Integrated Approach.
8. ASM Handbook for Material Properties.
9. Automotive Chassis Engineering by Reimpell and Stoll.
10. Standard Design Data Handbook for Mechanical Engineers.

SYMBOLS USED

Symbol	Description
W	Total load
M	Bending moment
V	Shear force
σ	Normal stress
τ	Shear stress
I	Moment of inertia
D	Outer diameter
d	Inner diameter
FOS	Factor of safety

FUTURE RECOMMENDATIONS

Future improvements may include:

- Finite element analysis using ANSYS.
- Fatigue life estimation.
- Experimental testing.
- Dynamic vibration analysis.
- Composite material axle design.
- Corrosion protection studies.

MOM-II

Given Data:

Vehicle Load: mass $m = 1500 \text{ kg}$

Length: $L = 1.6 \text{ m}$

Material Selected: Medium Carbon Steel
(SAE 4140)

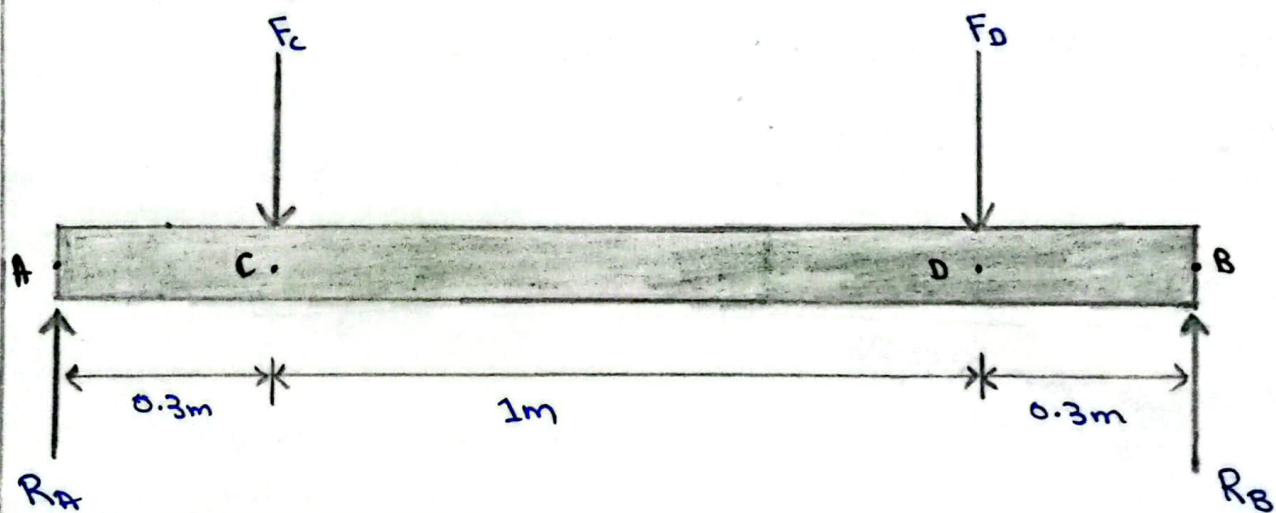
Yield Strength: 415 MPa (Normal)

Yield Strength: 240 MPa (Shear)

Rear dead axle load assumption: 65%

Factor of Safety: 2.5 (assumed)

Free Body Diagram:



To Calculate F_c & F_d :

$$F_c = F_d = \frac{1500}{2} \times 9.81 \times (0.65)$$

$$\boxed{F_c = F_d = 4782.375 \text{ N}}$$

Reaction Forces:

$$\sum F_y = 0 \quad +\uparrow$$

$$R_A - F_c - F_d + R_B = 0$$

$$R_A + R_B = 4782.375 \times 2$$

$$R_A + R_B = 9564.75 \text{ N} \quad \text{--- (1)}$$

$$\sum M_A = 0 \quad (+)$$

$$-4782.375(0.3) - 4782.375(1.3) + R_B(1.6) = 0$$

$$R_B = \frac{4782.375(0.3+1.3)}{1.6}$$

$$R_B = \frac{4782.375 \times (1.6)}{1.6}$$

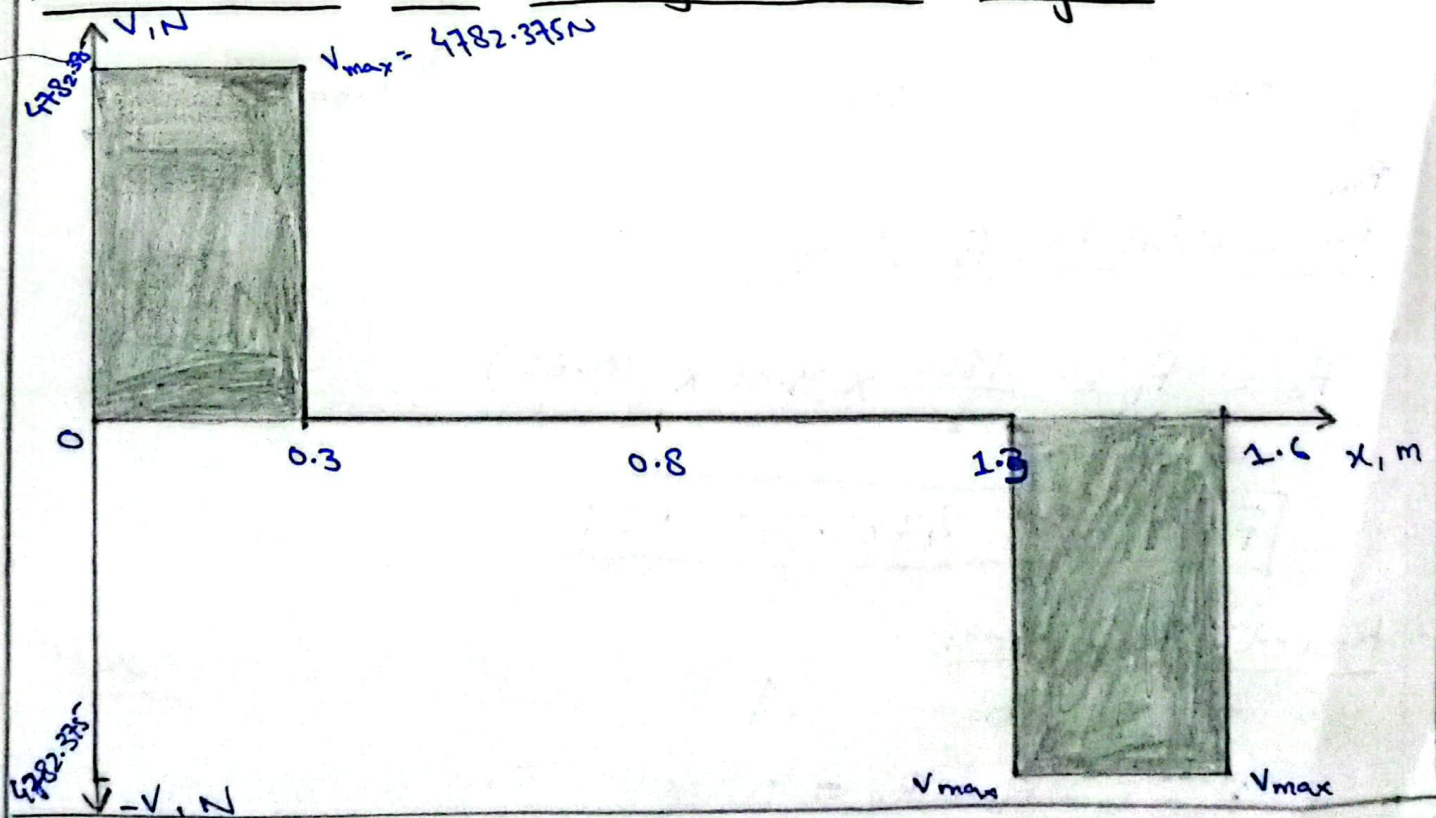
$$R_B = 4782.375 \text{ N}$$

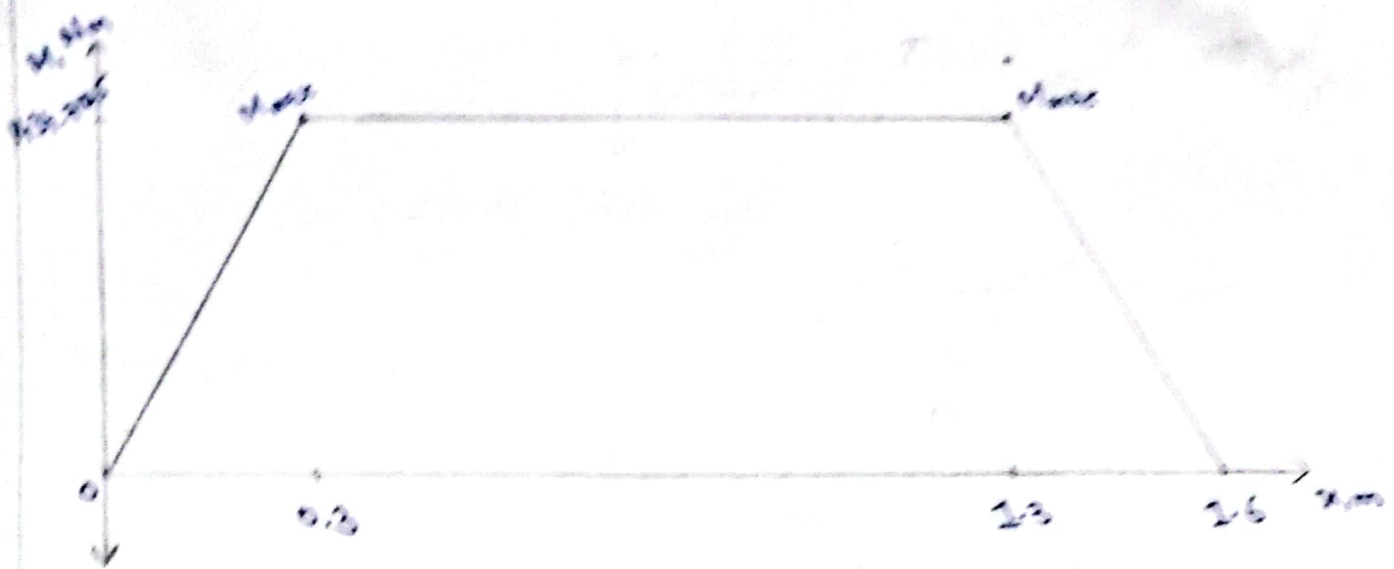
Put in (1), we get

$$R_A = 9564.75 - 4782.375$$

$$R_A = 4782.375 \text{ N}$$

Shear Force And Bending Moment Diagram:





$$\left. \begin{aligned} V_{\max} &= 4782.375 \text{ N} \\ M_{\max} &= 1434.7125 \text{ Nm} \end{aligned} \right\} \text{ at point C}$$

Allowable Stresses:

$$F.O.S = \frac{\text{Yield Strength}}{\text{allowable stress}}$$

$$\rightarrow \text{Allowable Stress} = \frac{\text{Yield Strength (normal)}}{F.O.S}$$

$$\text{Allowable Stress} = \frac{415}{2.5}$$

$$\text{Allowable Stress} = 166 \text{ MPa (Normal)}$$

$$\rightarrow \text{Allowable stress} = \frac{\text{Yield Strength (Shearing)}}{F.O.S}$$

$$\text{Allowable Stress} = \frac{240}{2.5}$$

$$\text{Allowable Stress} = 96 \text{ MPa}$$

Using Bending Stress Formula:

$$\sigma_{all} = \frac{Mc}{I}$$

$$c = \frac{d}{2}, \quad I = \frac{\pi d^4}{64}$$

For Solid shaft

$$\tau_{all} = \frac{M_{max} \cdot d/2}{\frac{\pi d^4}{64}}$$

$$\tau_{all} = \frac{32 M_{max}}{\pi d^3}$$

$$d^3 = \frac{32 M_{max}}{\pi \tau_{all}}$$

$$d = \sqrt[3]{\frac{32 M_{max}}{\pi \tau_{all}}}$$

$$d = \sqrt[3]{\frac{32 \times 1434.7125 \text{ Nm}}{\pi \times 166 \times 10^6 \text{ Nm}^{-2}}}$$

$$d = 0.044486 \text{ m}$$

$$\boxed{d \approx 45 \text{ mm}} \rightarrow \text{approximately } 45 \text{ mm}$$

Maximum Normal stress:

$$\tau_{max} = \frac{M_{max} c}{I}$$

$$\tau_{max} = \frac{32 M_{max}}{\pi d^3}$$

$$\tau_{max} = \frac{32 \times 1434.7125}{\pi \times (0.045)^3}$$

$$\tau_{max} = 160371594.2 \text{ Pa}$$

$$\sigma_{\max} = 160.4 \text{ MPa}$$

$$160.4 \text{ MPa} < 166 \text{ MPa}$$

$\sigma_{\max} < \sigma_{\text{an}}$ (it will not fail at $d = 45 \text{ mm}$)

Maximum Shearing Stress:

Breaking Force on each wheel

$$F_B = 4782.375 \text{ N} \times \mu_{\text{road}}$$

$$F_B = 4782.375 \times 0.7$$

$$F_B = 3347.6625 \text{ N}$$

Torque Produced:

$$T = F_B \times r_{\text{wheel}}$$

[Selecting wheel radius $r = 300 \text{ mm}$]

$$T = F_B \times r_{\text{wheel}}$$

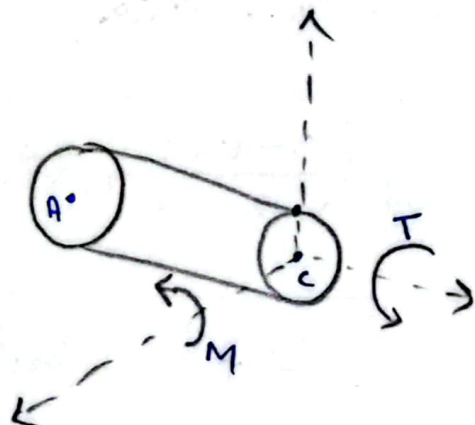
$$T = 3347.6625 \text{ N} \times 0.3 \text{ m}$$

$$T = 1004.29875 \text{ Nm}$$

$$\tau_m = \frac{Tc}{J}$$

$$\tau_m = \frac{T \times d/2}{\pi d^4 / 32}$$

$$\tau_m = \frac{T}{\pi d^3 / 16}$$



$$\tau_m = \frac{16 T}{\pi d^3}$$

$$\tau_m = \frac{16 \times 1004.29875}{\pi \times (0.045)^3}$$

$$\tau_m = 56130057.97$$

$$\tau_m = 56.1 \text{ MPa}$$

$$56.1 \text{ MPa} < 96 \text{ MPa}$$

$\tau < \tau_{\text{all}}$ (It will not fail)

Principal Stresses at Point C:

$$\sigma_m = -160.4 \text{ MPa}$$

$$\tau_m = 56.1 \text{ MPa}$$

$$\sigma_{\text{avg}} = \frac{\sigma_m + 0}{2}$$

$$\sigma_{\text{avg}} = \frac{-160.4 + 0}{2}$$

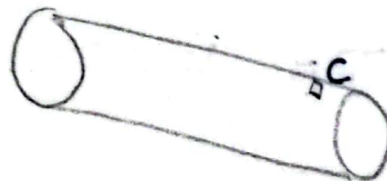
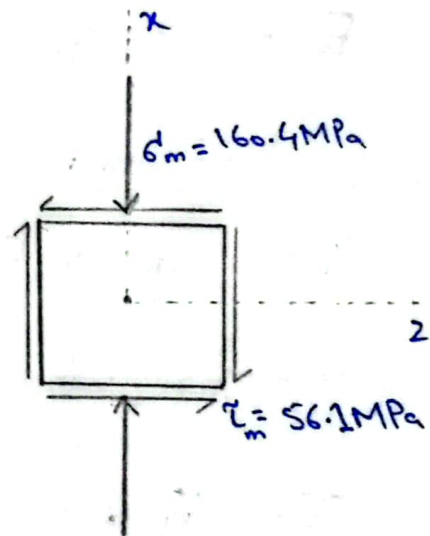
$$\sigma_{\text{avg}} = -80.2 \text{ MPa}$$

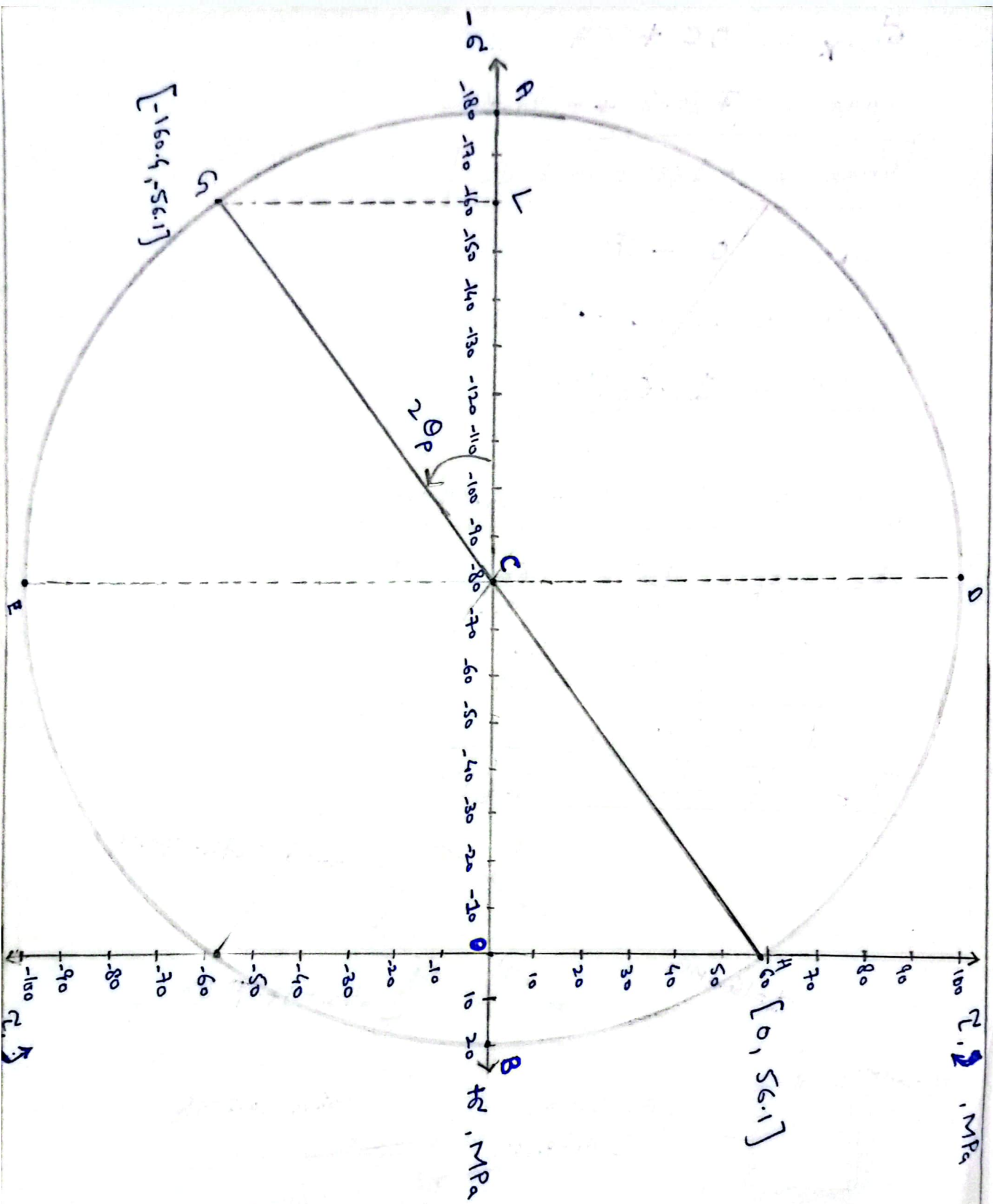
$$R = \sqrt{\left(\frac{0 - \sigma_m}{2}\right)^2 + (\tau_m)^2}$$

$$R = \sqrt{(80.2)^2 + (56.1)^2}$$

$$R = 97.87 \text{ MPa}$$

Drawing Mohr's Circle for it:





Scale $\downarrow$
 $1\text{cm} = 10\text{MPa}$

$$\sigma_{\max} = OC + CA$$

$$\sigma_{\max} = 780.2 - 97.87$$

$$\sigma_{\max} = -178.07 \text{ MPa}$$

$$\sigma_{\min} = OC - CB$$

$$\sigma_{\min} = -80.2 - (-97.87)$$

$$\sigma_{\min} = 17.67 \text{ MPa}$$

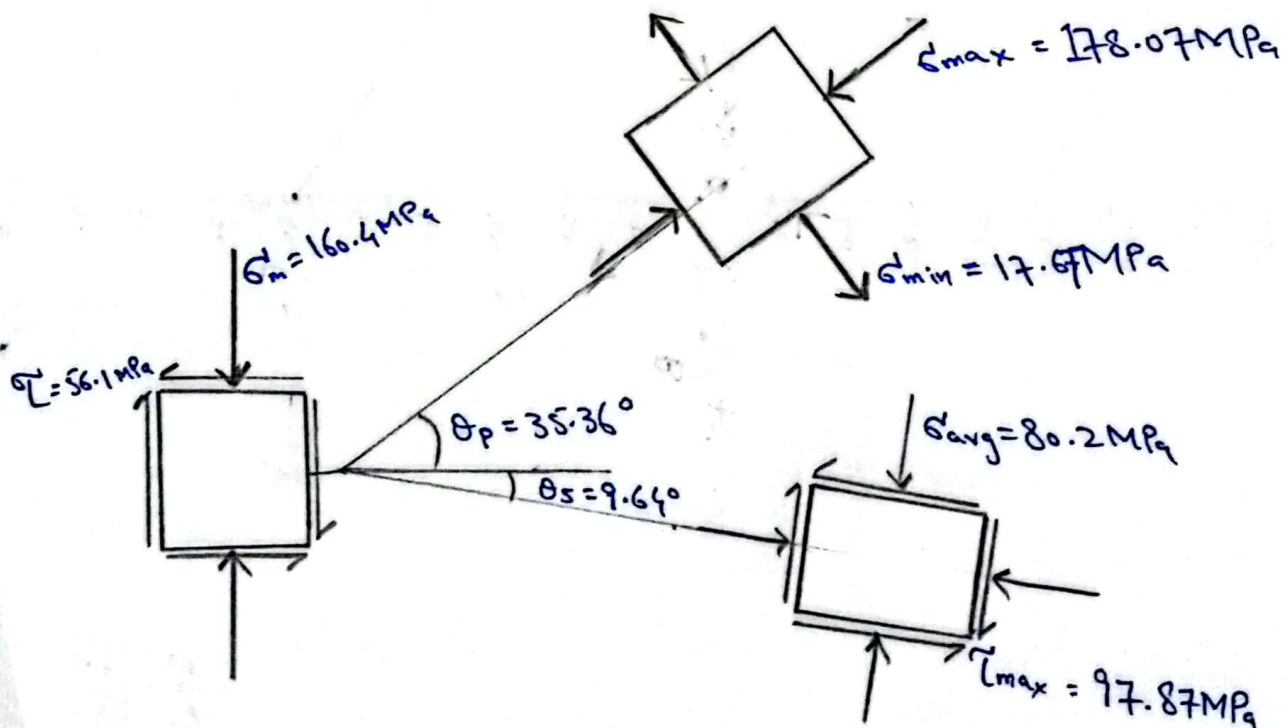
$$\tau_{\max} = CD = R = 97.87 \text{ MPa}$$

$$\tan 2\theta_p = \frac{LG}{CL} = \frac{160.4}{56.1}$$

$$\theta_p = \frac{1}{2} \tan^{-1} \left(\frac{160.4}{56.1} \right)$$

$$\theta_p = 35.36^\circ$$

$$\theta_s = 45 - 35.36 = 9.64^\circ$$



The factor of safety we assumed 2.5 is not suitable for this applied load, if we reduce FOS from 2.5 to 2.3 then beam will not fail, with for that $d = 45\text{mm}$.

FOS must be less than 2.5

Assume 2.3 now,

$$\sigma_{all} = \frac{415}{2.3} = 180.4 \text{ MPa}$$

$$\tau_{all} = \frac{240}{2.3} = 104.3 \text{ MPa}$$

now,

$$\sigma_{all} = \frac{Mc}{I}$$

re arranging for d ,

$$d = \sqrt[3]{\frac{32M}{\pi \sigma_{all}}}$$

$$d = \sqrt[3]{\frac{32 \times 1434.7125}{\pi \times 180.4 \times 10^6}}$$

$$d = 0.04326 \text{ m}$$

$$d = 43.3 \text{ mm or } \boxed{d \approx 45 \text{ mm}}$$

for $d = 45\text{mm}$

$$\sigma_{max} = -178.07 \text{ MPa}$$

$$\text{or } |\sigma_{max}| = 178.07 \text{ MPa}$$

$$|\sigma_{max}| \leq 180.4 \text{ MPa} \longrightarrow \text{Not Fail}$$

$$\tau_{\max} = 97.87 \text{ MPa}$$

$$\tau_{\max} \leq 104.3 \text{ MPa} \longrightarrow \text{Not fail}$$

Now Factor of Safety is 2.3